



Welcome to the Else Kooi Laboratory

February 2023 - HBO course version



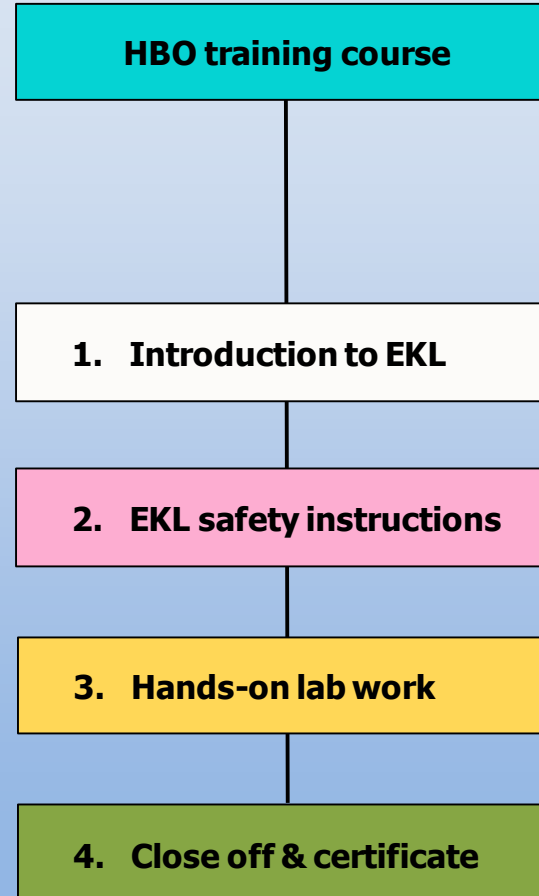
Introduction to the



EKL laboratories

(formerly known as the Dimes Technology Center)

Program for today



Introduction to EKL

Else Kooi Laboratory



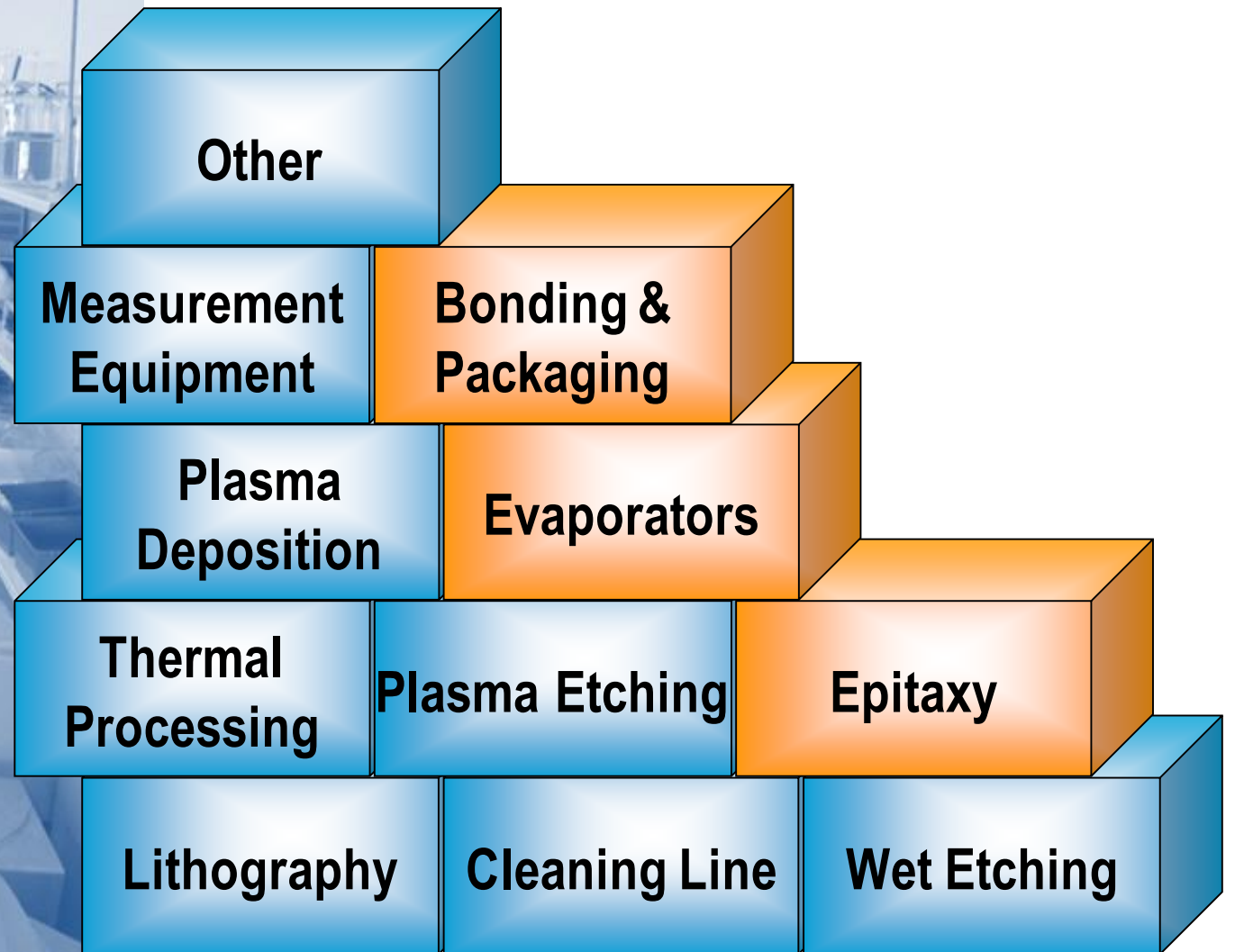
- EKL takes its name from the Dutch chemist [Else Kooi](#) (1932-2001).
- Main objectives:
 - Facilitate excellent (sub) microfabrication.
 - Support Science & Research as well as Prototyping & Small scale production.
 - Build a bridge between academic research and industrial innovation in the microfabrication field.



Else Kooi

EKL Processes

**and
more**

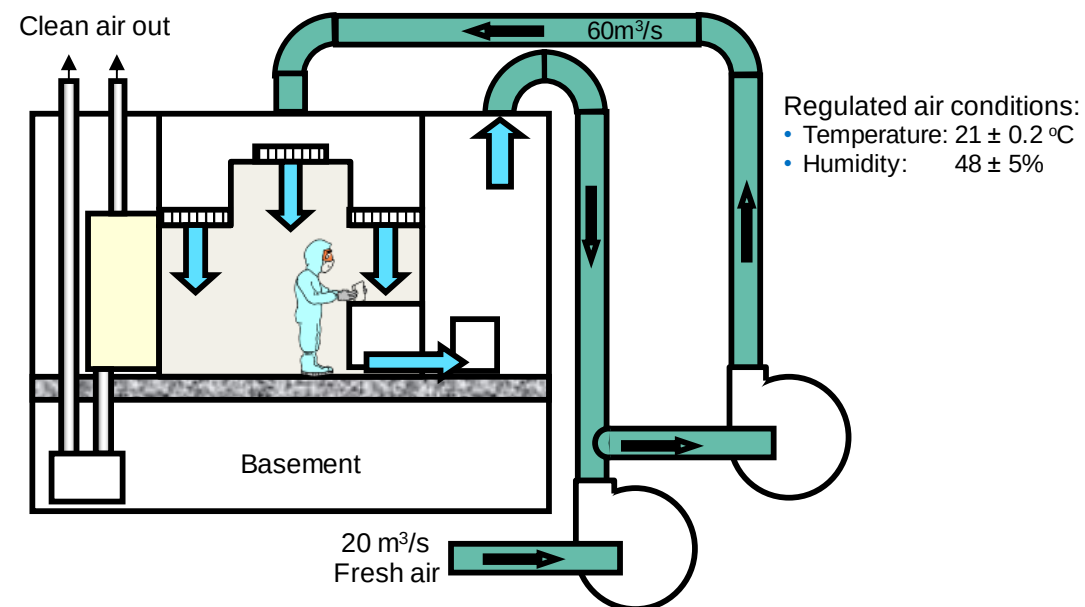


Cleanroom

A **clean room** is what your mother always wanted from you.



And a **cleanroom** is what we are discussing today.



Cleanroom definition

A cleanroom is defined as an enclosed area that is specially designed to control, monitor and maintain several parameters such as:

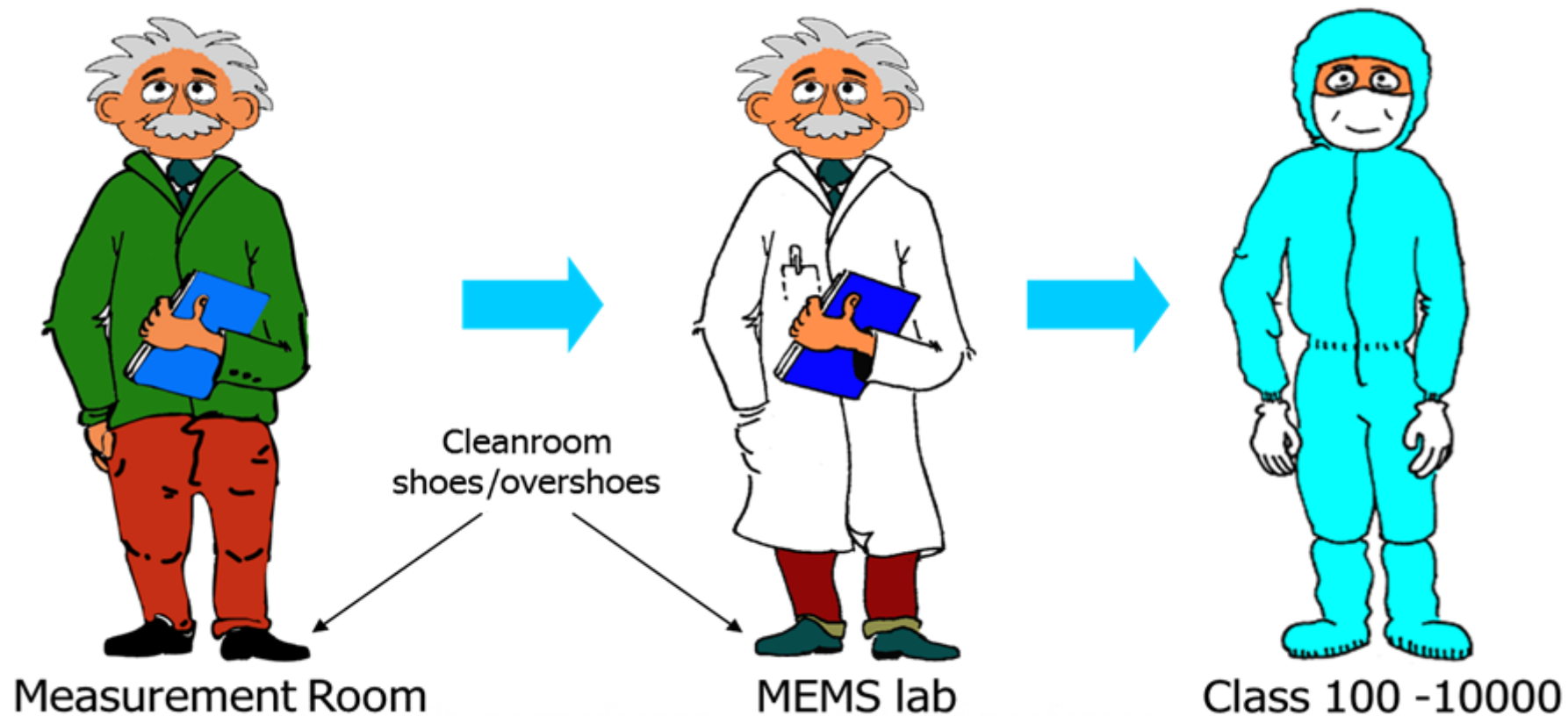
- air pollutant levels
- room pressure, temperature & humidity
- personnel access

to meet the critical environmental conditions that are demanded.

it's a

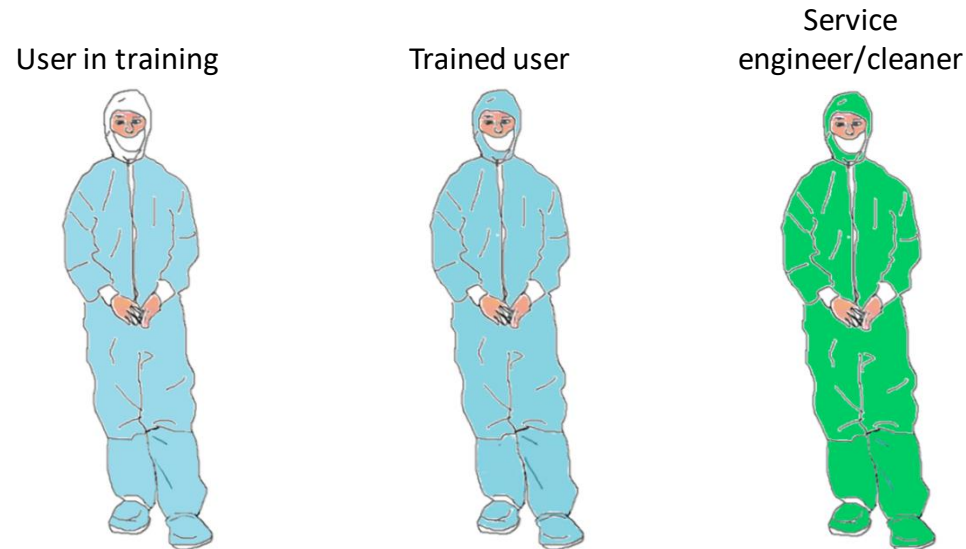
CONTROLLED ENVIRONMENT

EKL lab clothing policy



Gowning / de-gowning procedure

- In the cleanroom all users must wear a **cleanroom suit** at all times. These suits include: coveralls, a hood, a mouth cover, boots and gloves.
- Please look at the video film for the correct gowning procedure
<https://www.youtube.com/watch?v=IJSUUt看2wds>
- The instructions for wearing the cleanroom garments in order to control particle contamination is also posted on the **graphical guide** in the cleanroom gowning room.



Some cleanroom rules

- Once you are dressed in a cleanroom suit and have entered the lab, you should **not unzip** the suit.
- If you need something from your pocket, you must return to the gowning area and retrieve it there.
- Items that are taken into any cleanroom must be wiped down with **PurWipe microfiber wipes**. These wipes are moisturized with a water/IPA solution, and can be found in the gowning areas.
- Generating particles in a cleanroom must be minimized at all time.
Hence it is **not allowed** to run, use normal paper, tear up paper, use a pencil, etc...
- It is allowed to take a mobile phone into the lab, but only when it is sealed in a plastic pouch that can be found in the gowning area.
- Eating and drinking in the labs is not allowed.



EKL Safety Instructions

Safety is the responsibility of everyone, everywhere.



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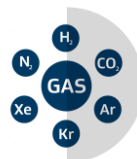
Safety risks

Working in a cleanroom or laboratory results in a higher safety risk than working in an office environment:

- **Liquids** and **solids** are being used.



- **Gases** are being used.



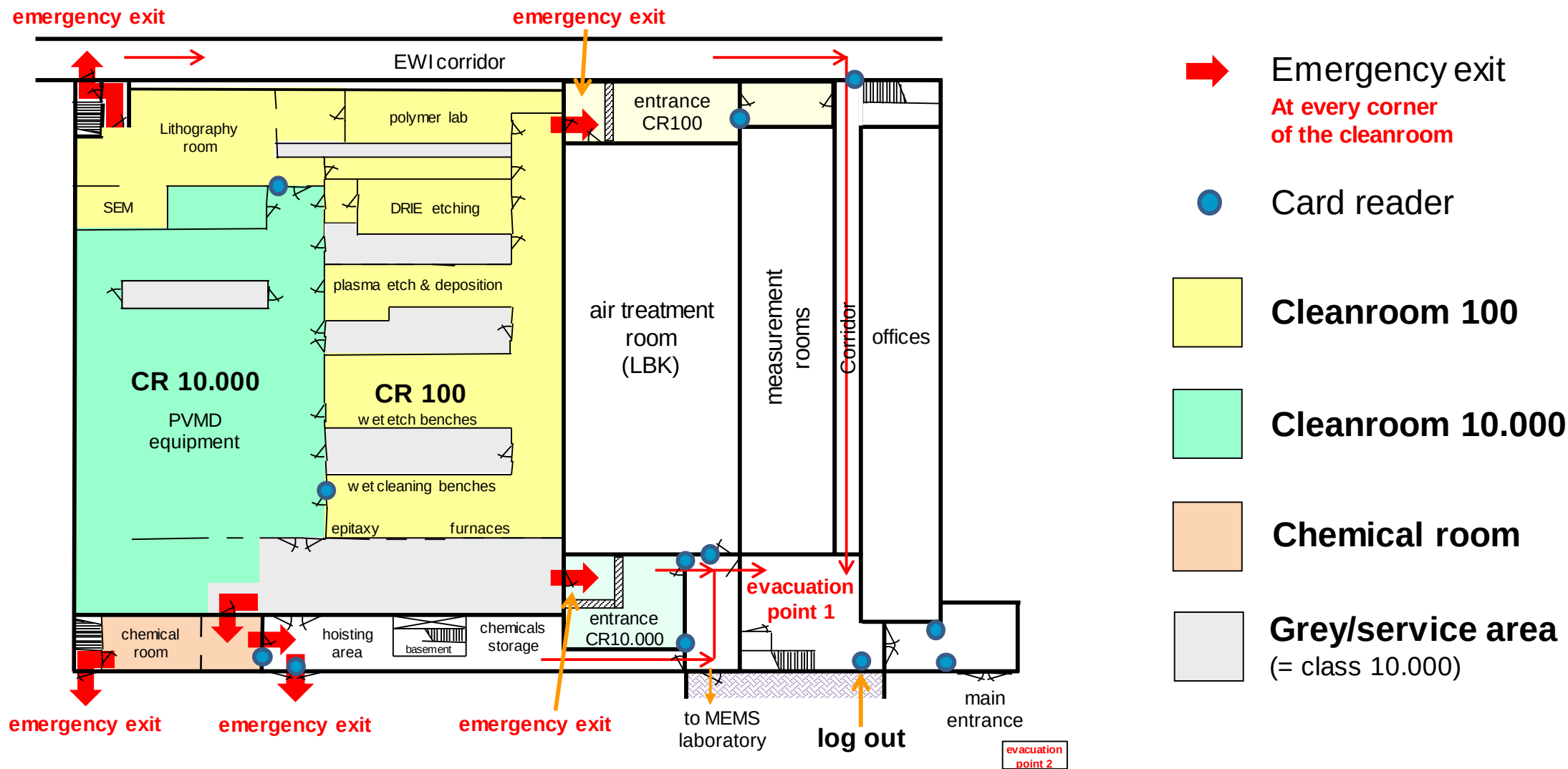
Some of these are **hazards** !!!

Requirements to work in the labs

- Become familiar with the **map of the labs**.
- **Know where to find:**
 - emergency exits
 - alarm buttons
 - fire extinguishers
 - emergency kits
 - showers
- **Keep access free to:**
 - escape routes
 - fire extinguishers
 - emergency kits
- Always follow the instructions of the **Emergency Response Team (ERT)**.



Cleanroom escape routes in case of an alarm



Access control

The access control system:

The main purpose of access control is to **provide security** to the EKL users and visitors by allowing or restricting access to the labs.

The following rules apply:

- Each lab user must use a **valid badge** to enter or exit a lab.
- It is **not allowed** that 2 or more users enter a lab with one badge.
- Visitors & guests must be escorted by a staff member or **qualified** user.

When entering or leaving a lab:

- **Sign in:** Use your **own** training badge to enter the room.
- **Log out:** Use your **own** training badge when leaving the room.

In case of an alarm or incident the Emergency Response Team and/or the Fire Brigade can check who is still where in the EKL building.



Type of alarms

- Automatic smoke detector alarm.
This activates an alarm sound, and on the alarm panels only the text "Cleanroom verlaten" (= leave cleanroom) will light up.
- Automatic gas detector alarm.
This activates an alarm sound, and on the alarm panels both the text "Gasalarm" and "Cleanroom verlaten" will light up.
- Manual alarm (pressed button).
This activates an alarm sound, and on the alarm panels only the text "Cleanroom verlaten" will light up.



During alarm: Sound + Panel illuminated.

An alarm test is performed every first Monday of the month at 8:30 hrs.

What to do if there is an alarm

- Leave the work area immediately. Do not finish your process.
- Alert other users on your way out.
- Follow the escape route to the nearest emergency exit.
 - All doors will automatically unlock.
(If not, then press the button in the green box next to the door.)
 - Close the doors behind you.
 - Do **NOT** take off your cleanroom suit or lab coat.
- Go to evacuation point 1 and log out.
 - Here users can take off their cleanroom suit or lab coat.
- Wait for further instructions.

Note: Evacuation point 2 is only used if point 1 is unsafe.

If you know what caused the alarm then inform the ERT.



Sign indicating the escape route



Green box to unlock the door
(if not automatically unlocked)



Evacuation point 1

inside the building at
the reception hall



Evacuation point 2

outside the building in front
of the main entrance

